

India Major Crops

- With a population of 1.36 billion India is the world's second most populous country.
- It is the seventh largest country in the world with an area of 3.288 million sq kms.
- It has a long coastline of over 7,500 kms.
- India is a diverse country where over 22 major languages and 415 dialects are spoken

- With the highest mountain range in the world, the Himalayas to its North, the Thar desert to its West, the Gangetic delta to its East and the Deccan Plateau in the South, the country is home to vast agro-ecological diversity.

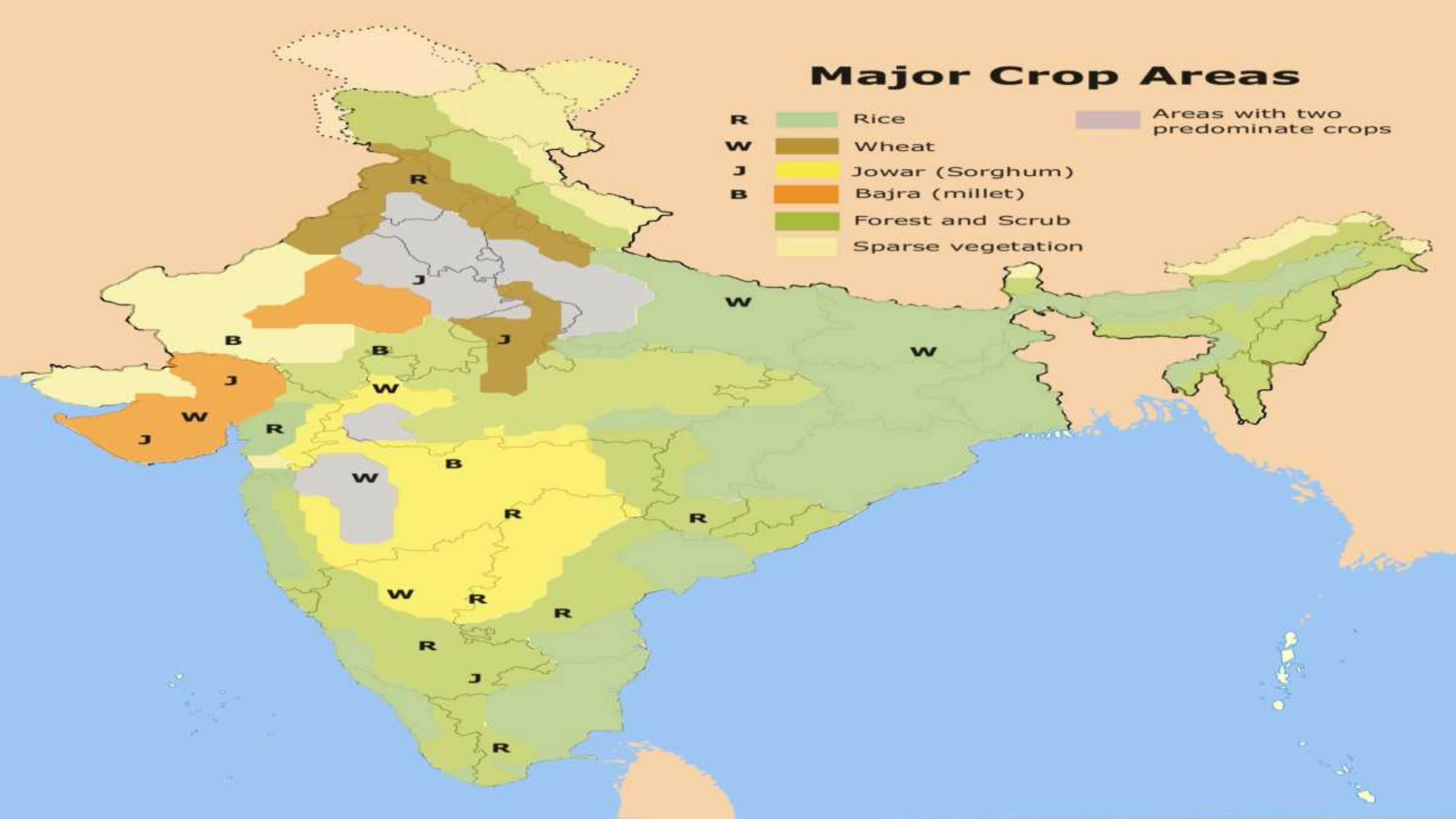
- India is the world's largest producer of milk, pulses and jute, and ranks as the second largest producer of rice, wheat, sugarcane, groundnut, vegetables, fruit and cotton. It is also one of the leading producers of spices, fish, poultry, livestock and plantation crops. Worth $ 2.1 trillion,India is the world's third largest economy after the US and China

- India's climate varies from humid and dry tropical in the south to temperate alpine in the northern reaches and has a great diversity of ecosystems. Four out of the 34 global biodiversity hotspots and 15 WWF global 200 eco-regions fall fully or partly within India. Having only 2.4 percent of the world's land area, India harbours around eight percent of all recorded species, including over 45,000 plant and 91,000 animal species.

- Agriculture, with its allied sectors, is the largest source of livelihood in India. 70 percent of its rural households still depend primarily on agriculture for their livelihood, with 82 percent of farmers being small and marginal. In 2017-18, total food grain production was estimated at 275 million tonnes (MT).

- India is the largest producer (25% of global production), consumer (27% of world consumption) and importer (14%) of pulses in the world. India's annual milk production was 165 MT (2017-18), making India the largest producer of milk, jute and pulses, and with world's second-largest cattle population 190 million in 2012.

- It is the second-largest producer of rice, wheat, sugarcane, cotton and groundnuts, as well as the second-largest fruit and vegetable producer, accounting for 10.9% and 8.6% of the world fruit and vegetable production, respectively.

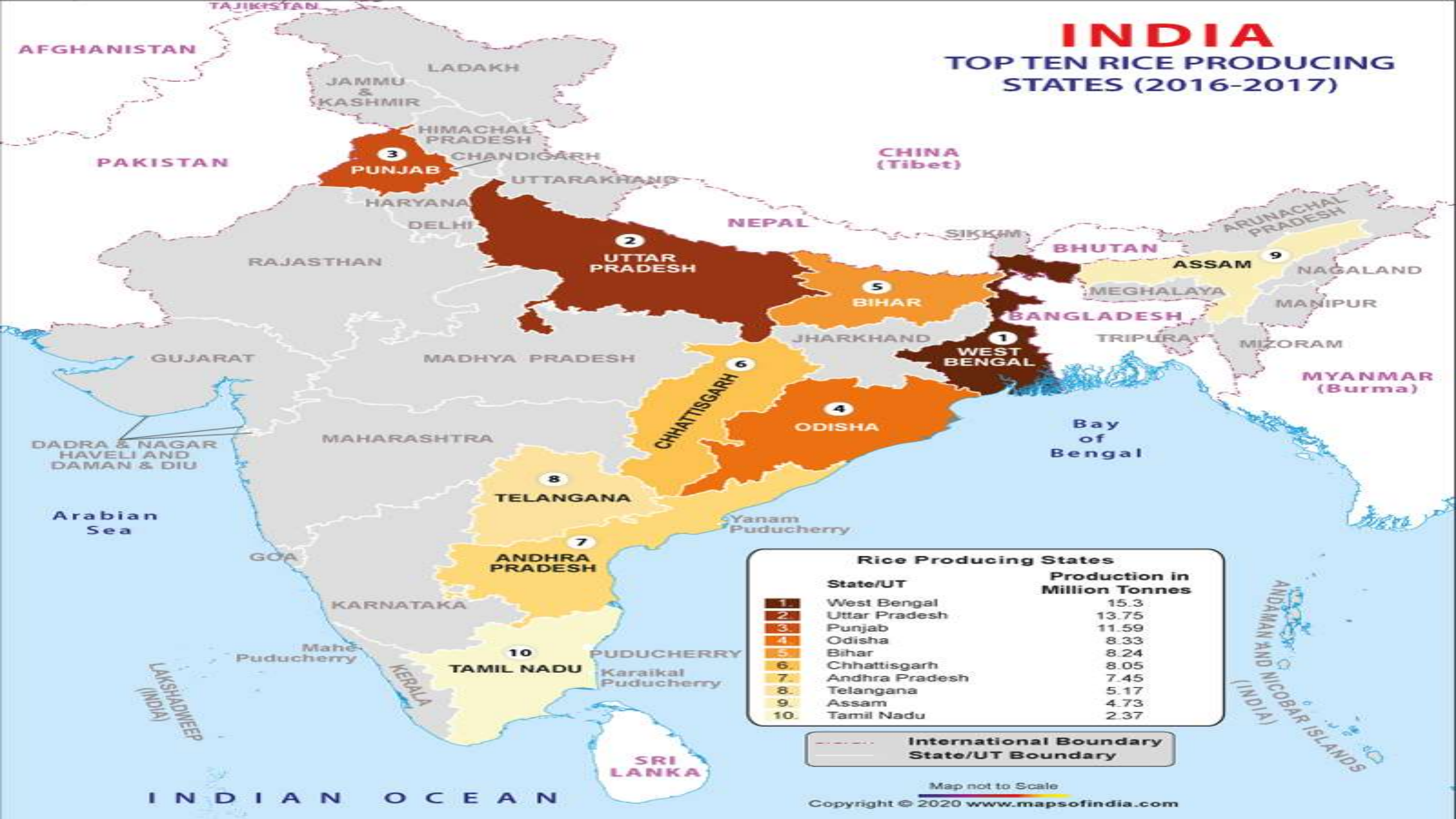

Major Crop Areas
R
Rice
W
Wheat
J
Jowar (Sorghum)
B
Bajra (millet)
Forest and Scrub
Sparse vegetation
Areas with two predominate crops

S. No	Cropping Season	Time Period	Crops	States
1.	Rabi	Sown: October-December Harvested: April-June	Wheat, barley, peas, gram, mustard etc.	Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand and Uttar Pradesh
2.	Kharif	Sown: June-July Harvested: September-October	Rice, maize, jowar, bajra, tur, moong, urad, cotton, jute, groundnut, soybean etc.	Assam, West Bengal, coastal regions of Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala and Maharashtra
3.	Zaid	Sown and harvested: March-July (between Rabi and Kharif)	Seasonal fruits, vegetables, fodder crops etc.	Most of the northern and northwestern states

Major Food Crops

- **Rice**
- Temperature: Between 22-32°C with high humidity.
- Rainfall: Around 150-300 cm.
- Soil Type: Deep clayey and loamy soil.
- Top Rice Producing States: West Bengal > Punjab > Uttar Pradesh > Andhra Pradesh > Bihar.

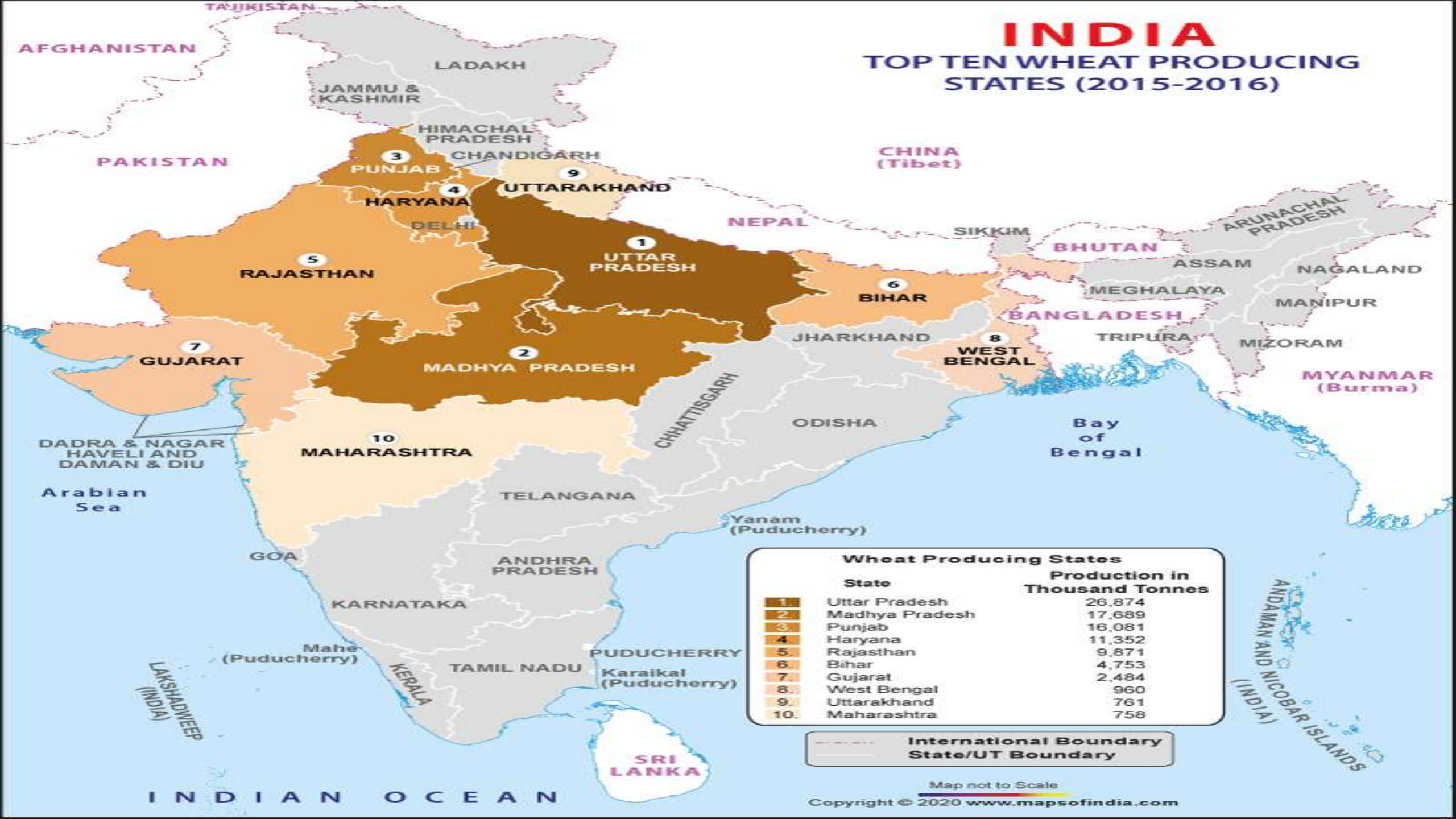
- It is the staple food crop of majority of Indian people.
- India is the second largest producer of rice in the world after China.
- In states like Assam, West Bengal and Odisha, three crops of paddy are grown in a year. These are Aus, Aman and Boro.
- National Food Security Mission, Hybrid Rice Seed Production and Rashtriya Krishi Vikas Yojana are few government initiatives to support rice cultivation.


INDIA
TOP TEN RICE PRODUCING STATES (2016-2017)
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (Tibet)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (Burma)
LADAKH
JAMMU & KASHMIR
HIMACHAL PRADESH
CHANDIGARH
3 PUNJAB
UTTARAKHAND
HARYANA
DELHI
2 UTTAR PRADESH
RAJASTHAN
SIKKIM
ARUNACHAL PRADESH
9 ASSAM
NAGALAND
MEGHALAYA
MANIPUR
5 BIHAR
JHARKHAND
TRIPURA
MIZORAM
1 WEST BENGAL
GUJARAT
MADHYA PRADESH
6 CHHATTISGARH
4 ODISHA
Bay of Bengal
DADRA & NAGAR HAVELI AND DAMAN & DIU
MAHARASHTRA
8 TELANGANA
Yanam Puducherry
Arabian Sea
7 ANDHRA PRADESH
GOA
KARNATAKA
Mahe Puducherry
10 TAMIL NADU
PUDUCHERRY
Karaikal Puducherry
KERALA
LAKSHADWEEP (INDIA)
ANDAMAN AND NICOBAR ISLANDS (INDIA)
SRI LANKA
INDIAN OCEAN
Rice Producing States
State/UT
Production in Million Tonnes
1. West Bengal 15.3
2. Uttar Pradesh 13.75
3. Punjab 11.59
4. Odisha 8.33
5. Bihar 8.24
6. Chhattisgarh 8.05
7. Andhra Pradesh 7.45
8. Telangana 5.17
9. Assam 4.73
10. Tamil Nadu 2.37
International Boundary
State/UT Boundary
Map not to Scale
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- **Wheat**
- Temperature: Between 10-15°C (Sowing time) and 21-26°C (Ripening & Harvesting) with bright sunlight.
- Rainfall: Around 75-100 cm.
- Soil Type: Well-drained fertile loamy and clayey loamy (Ganga-Satluj plains and black soil region of the Deccan)

- Top Wheat Producing States: Uttar Pradesh > Punjab > Madhya Pradesh > Haryana > Rajasthan.
- India is the second largest producer after China.
- This is the second most important cereal crop and the main food crop, in north and north-western India.

- Success of Green Revolution contributed to the growth of Rabi crops especially wheat.
- Macro Management Mode of Agriculture, National Food Security Mission and Rashtriya Krishi Vikas Yojana are few government initiatives to support wheat cultivation.


INDIA
TOP TEN WHEAT PRODUCING STATES (2015-2016)
TAJIKISTAN
AFGHANISTAN
PAKISTAN
LADAKH
JAMMU & KASHMIR
HIMACHAL PRADESH
CHANDIGARH
3 PUNJAB
9 UTTARAKHAND
4 HARYANA
DELHI
CHINA (Tibet)
NEPAL
SIKKIM
BHUTAN
ARUNACHAL PRADESH
ASSAM
NAGALAND
MEGHALAYA
MANIPUR
BANGLADESH
TRIPURA
MIZORAM
MYANMAR (Burma)
1 UTTAR PRADESH
5 RAJASTHAN
6 BIHAR
JHARKHAND
8 WEST BENGAL
7 GUJARAT
2 MADHYA PRADESH
CHHATTISGARH
ODISHA
Bay of Bengal
DADRA & NAGAR HAVELI AND DAMAN & DIU
10 MAHARASHTRA
Arabian Sea
TELANGANA
Yanam (Puducherry)
GOA
ANDHRA PRADESH
KARNATAKA
Mahe (Puducherry)
PUDUCHERRY
Karaikal (Puducherry)
TAMIL NADU
KERALA
LAKSHADWEEP (INDIA)
SRI LANKA
INDIAN OCEAN
ANDAMAN AND NICOBAR ISLANDS (INDIA)
Wheat Producing States
State
Production in Thousand Tonnes
1. Uttar Pradesh 26,874
2. Madhya Pradesh 17,689
3. Punjab 16,081
4. Haryana 11,352
5. Rajasthan 9,871
6. Bihar 4,753
7. Gujarat 2,484
8. West Bengal 960
9. Uttarakhand 761
10. Maharashtra 758
International Boundary
State/UT Boundary
Map not to Scale
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- **Millets (Nutri-Cereals)**
- Temperature: Between 27-32°C
- Rainfall: Around 50-100 cm.
- Soil Type: Can be grown in inferior alluvial or loamy soil because they are less sensitive to soil deficiencies.

- Jowar- Rain-fed crop grown in the moist areas with less or no irrigation.
- Bajra- Sandy soils and shallow black soil.
- Ragi- Red, black, sandy, loamy and shallow black soils. (dry regions)

- Top Millets Producing States: Rajasthan > Karnataka > Maharashtra > Madhya Pradesh > Uttar Pradesh
- Jowar: Maharashtra > Karnataka > Madhya Pradesh > Tamil Nadu > Andhra Pradesh.
- Bajra: Rajasthan > Uttar Pradesh > Gujarat > Madhya Pradesh > Haryana.
- These are also known as coarse grains, which have high nutritional value. Ragi is very rich in iron, calcium, other micro nutrients and roughage.

- Jowar is the third most important food crop with respect to area and production.
- National Agricultural Insurance Scheme, Initiative for Nutritional Security through Intensive Millets Promotion are examples of government’s efforts to support millet production.

- **Maize**
- Temperature: Between 21-27°C
- Rainfall: High rainfall.
- Soil Type: Old alluvial soil.
- Top Maize Producing States: Karnataka > Maharashtra > Madhya Pradesh > Tamil Nadu > Telangana
- India is the seventh largest producer worldwide

- It is used both as food and fodder.
- Use of modern inputs such as High-Yielding Variety seeds, fertilisers and irrigation have contributed to the increasing production of maize.

- **Pulses**
- Temperature: Between 20-27°C
- Rainfall: Around 25-60 cm.
- Soil Type: Sandy-loamy soil.
- Top Pulses Producing States: Madhya Pradesh > Rajasthan > Maharashtra > Uttar Pradesh > Karnataka.

- India is the largest producer as well as the consumer of pulses in the world.
- These are the major source of protein in a vegetarian diet.
- Major pulses grown in India are tur (arhar), urad, moong, masur, peas and gram.

- Being leguminous crops, all these crops except arhar help in restoring soil fertility by fixing nitrogen from the air. Therefore, these are mostly grown in rotation with other crops.
- National Food Security Mission for Pulses, Pulses Development Scheme and Technological Mission on Pulses are few of the government’s plans to support pulses production.

Sugarcane

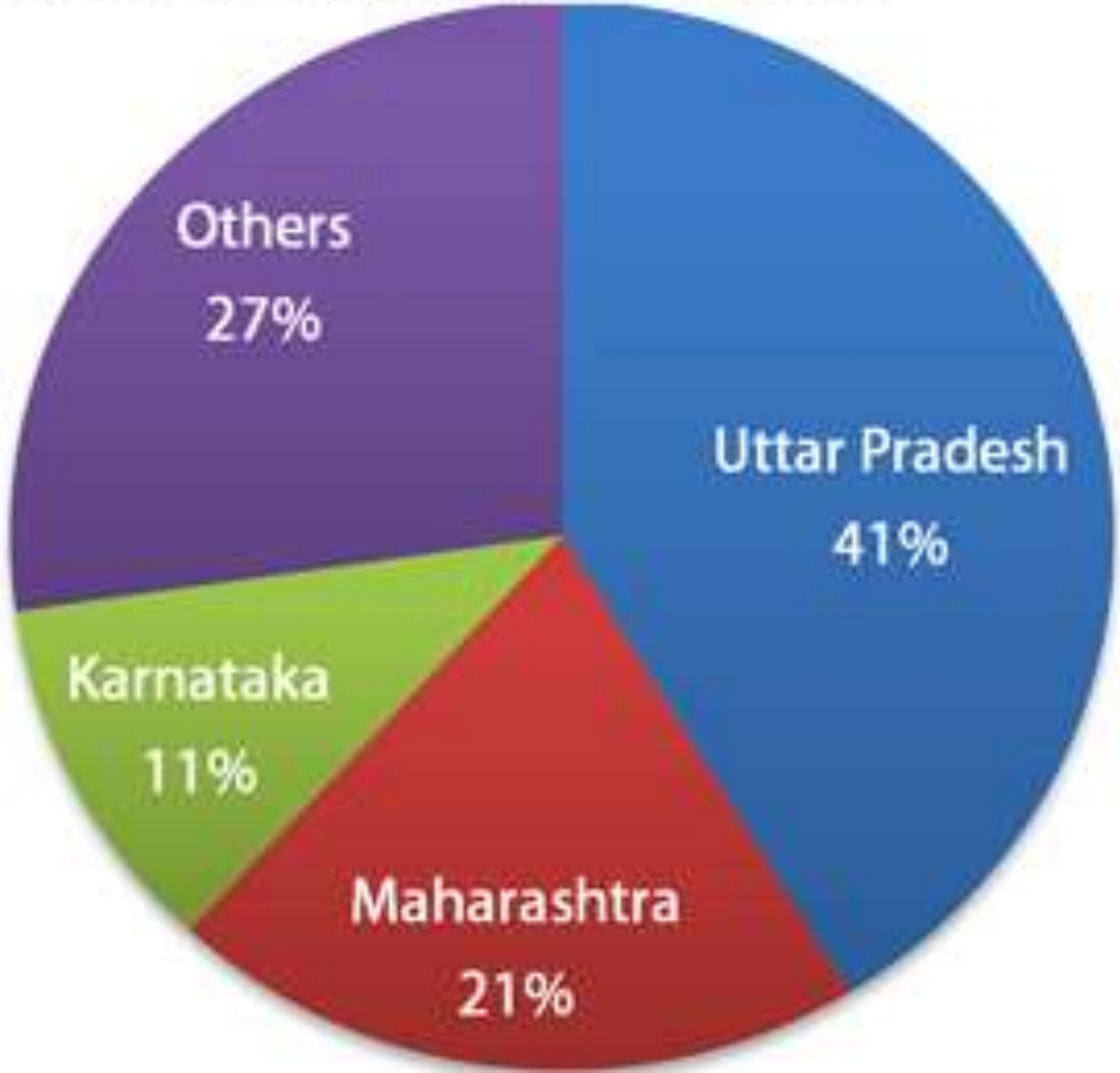
- In India, Sugarcane is grown as a Kharif Crop. It needs hot and humid climate with an average temperature of 21°C to 27°C. 75-150 cm rainfall is favorable for sugar cane cultivation

- Irrigation needed for areas with lesser rainfall. Sugarcane can grow in any soil which can retain moisture. Ideal soil for sugarcane is deep rich loamy soil. The soil needs to be rich in nitrogen, calcium and phosphorus but neither it should be neither too acidic nor too alkaline.

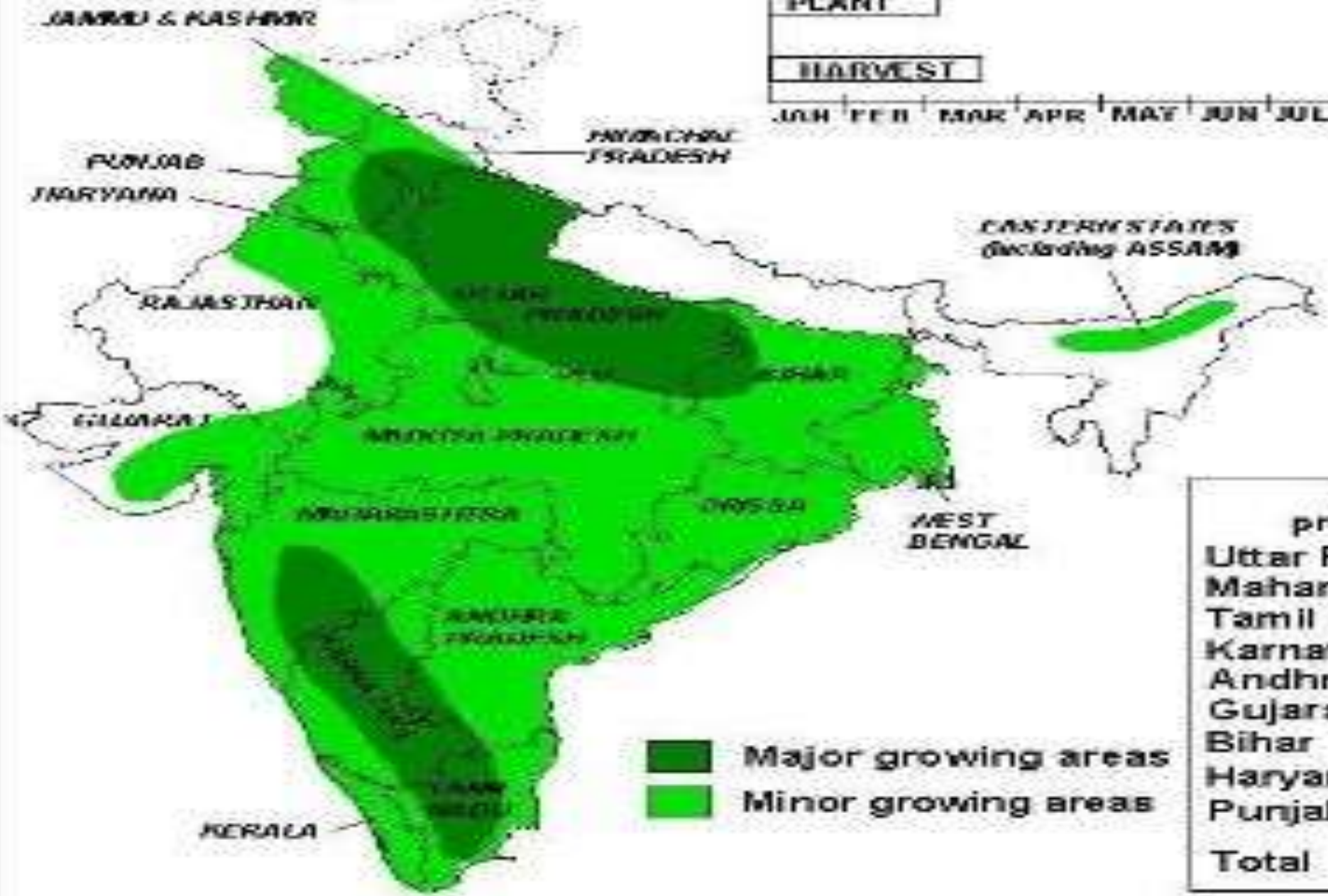
- Sugarcane is grown on around 2.8% of Gross Cropped Area of India. India produced around 352 million tonnes of sugar in 2015-16. Largest sugarcane producing state of India is Uttar Pradesh, which has 38.61% share in overall sugarcane production as per 2018-19 figures..

Largest Sugarcane Producing States of India

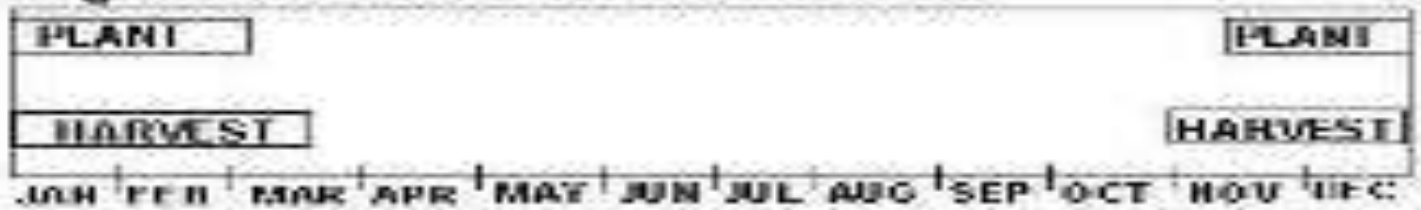
Uttar Pradesh	145.39
Maharashtra	72.26
Karnataka	38.48
All India	352.16



India: Sugarcane



Sugarcane calendar for most of India



Percent of total production by state	
Uttar Pradesh	44%
Maharashtra	14%
Tamil Nadu	10%
Karnataka	9%
Andhra Pradesh	5%
Gujarat	4%
Bihar	3%
Haryana	3%
Punjab	3%
Total	95%

JOINT AGRICULTURAL WEATHER FACILITY (USDA/NOAA)

- The second and third largest states are Maharashtra and Karnataka. Other main sugarcane producing states of India include Bihar, Assam, Haryana, Gujarat, Andhra Pradesh and Tamil Nadu

- Sugarcane is a water thirsty crop and enormous amount of water is needed in its production. The problem of optimal utilization of water becomes a major issue in those areas where irrigated crops and dryland farming are mixed..

- For example cultivation of irrigated crops such as rice/ sugarcane and dryland crops such as sorghum/millet on same watershed would result in a challenge to use the water optimally for both; because while former are water guzzling crops; too much water to the later is waste of this scarce resource

- In the conventional method, 2-3 budded sugarcane setts are used for planting. However, in SSI, they use single budded chips from a healthy mother cane. This would give high percentage of germination depending on the agro climatic conditions.
- Transplanting young seedlings
- The single budded chips raised are transplanted within 25-30 days in the main field.

- **Location of Production of Sugarcane**
- In India, sugarcane is produced in both tropical and subtropical regions. There are three distinct geographical regions in which sugarcane is produced.

- Sutlej-Ganga plain from Punjab to Bihar containing 51% of the total area and 60% of the country’s total production.
- The black soil belt from Maharashtra to Tamil Nadu along the eastern slopes of the Western Ghats.
- Coastal Andhra Pradesh and Krishna river valley.

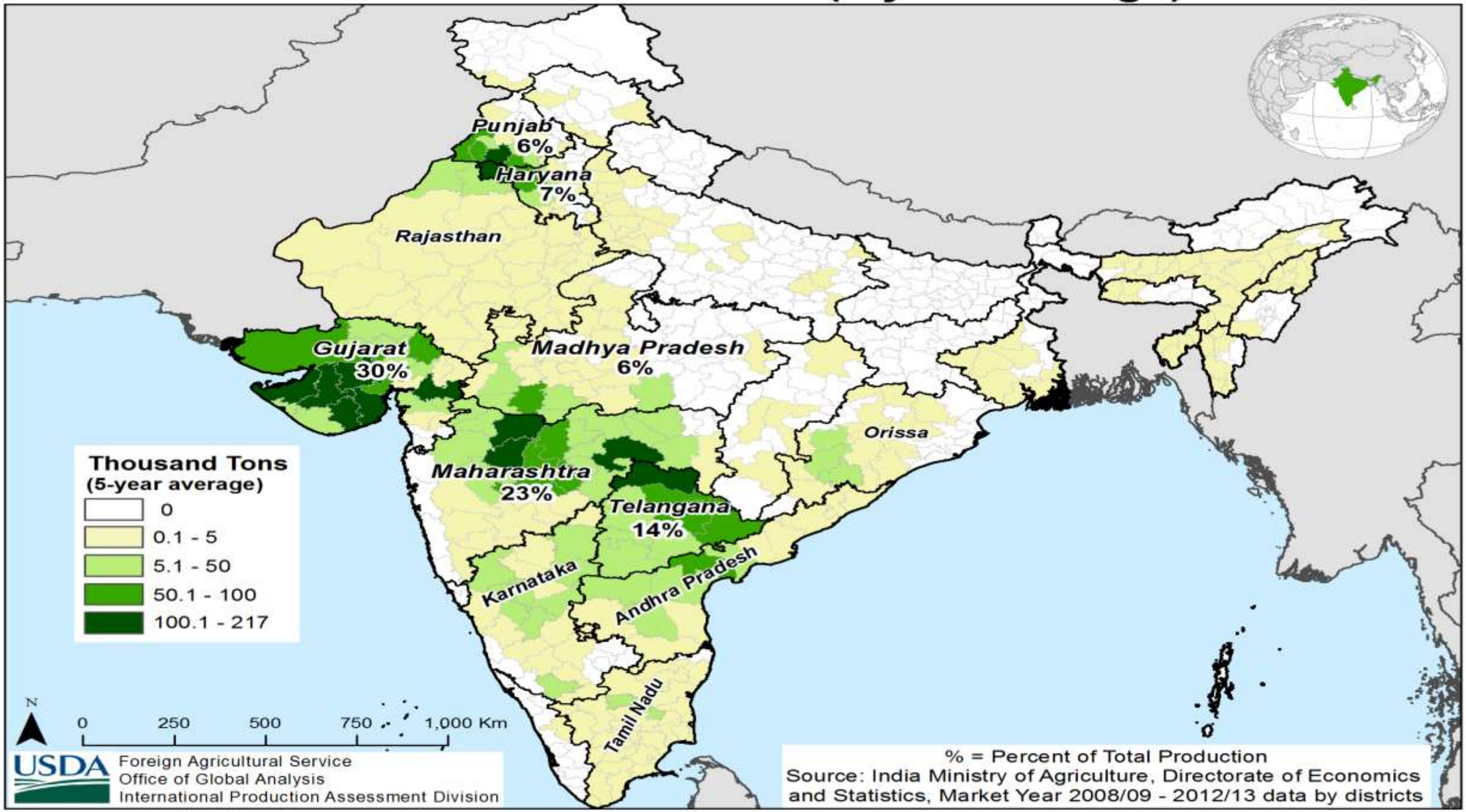
Cotton

- Cotton is the most important fibre crop not only of India but of the entire world. It provides the basic raw material (cotton fibre) to cotton textile industry.

- Cotton is the crop of tropical and sub-tropical areas and requires uniformly high temperature varying between 21°C and 30°C. The growth of cotton is retarded when the temperature falls below 20°C. Frost is enemy number one of the cotton plant and it is grown in areas having at least 210 frost free days in a year.

- The modest requirement of water can be met by an average annual rainfall of 50- 100 cm. However, it is successfully grown in areas of lesser rainfall with the help of irrigation.

India Cotton Production (5-year Average)



- About one-third of the total area under cotton cultivation is irrigated. In the year 2018-19 an area of 24 77 lakh hectares out of a total of 73.43 lakh hectares i.e. 33.73 per cent of the total area under cotton was irrigated

- Cotton is a kharif crop which requires 6 to 8 months to mature. Its time of sowing and harvesting differs in different parts of the country depending upon the climatic conditions. In Punjab and Haryana it is sown in April-May and is harvested in December-January that is before the winter frost can damage the crop.

- In the peninsular part of India, it is sown up to October and harvested between January and May because there is no danger of winter frost in these areas. In Tamil Nadu, it is grown both as a kharif and as a rabi crop.

- Cotton cultivation is closely related to deep black soils (regur) of the Deccan and the Malwa Plateaus and those of Gujarat. It also grows well in alluvial soils of the Satluj-Ganga Plain and red and laterite soils of the peninsular regions. Cotton quickly exhausts the fertility of soil. Therefore, regular application of manures and fertilizers to the soils is very necessary.

OIL SEEDS

Ground Nut

- 1. Gujarat (29.63%)
- 2. Tamil Nadu (20.78%)
- 3. Andhra Pradesh (15.23%)
- 4. Rajasthan (8.23%)
- 5. Maharashtra (8.23%)

Soyabean

- 1. Madhya Pradesh (54.01%)
- 2. Maharashtra (32.66%)
- 3. Rajasthan (8.70%)
- 4. Andhra Pradesh (1.81%)
- 5. Karnataka (1.02%)

- **Sunflower**
  - 1. Karnataka (42.28%)
  - 2. Andhra Pradesh (26.83%)
  - 3. Maharashtra (16.26%)
  - 4. Tamil Nadu (4.07%)
  - 5. Bihar (2.44%)

- **Jute**
- 1. West Bengal (75.51%)
- 2. Bihar (12.33%)
- 3. Assam (5.15%)
- 4. Andhra Pradesh (4.79%)
- 5. Orissa (1.15%)

- **Tobacco**
  - 1. Andhra Pradesh (35.04)
  - 2. Uttar Pradesh (24.07)
  - 3. Gujarat (12.93)
  - 4. Karnataka (8.86)
  - 5. Bihar (3.10)

- **Rubber:**
- 1. Kerala 2. Tamil Nadu
- 3. Karnataka 4. Andaman & Nicobar

- **Tea**
- 1. Assam (48%) 2. West Bengal
- 3. Tamil Nadu 4. Kerala

- **Coffee**

1.Karnataka (58%) 2. Kerala (24%)

3. Tamil Nadu 4. Andhra Pradesh

- **Spices**
- Chilli: 1. Andhra Pradesh 2. Karnataka 3. Maharashtra
- Turmeric: 1 Telangana 2. Tamil Nadu
- Cardamon: 1. Kerala 2. Karnataka
- Black Pepper: 1. Kerala 2. Karnataka
- Ginger: 1. Kerala 2. Meghalaya
- Kerala is called Spice Garden of India

- **Cashew**
  - 1. Maharashtra (31.77%)
  - 2. Andhra Pradesh (15.97%)
  - 3. Orissa (13.54%)
  - 4. Kerala (11.61%)
  - 5. Tamil Nadu (9.68%)

- **Onion**
- 1. Gujarat (29.73%)
- 2. Maharashtra (15.53%)
- 3. Karnataka (11.80)
- 4. Madhya Pradesh (8.62)
- 5. Andhra Pradesh (8.09)

- **Potato**
  - 1. Uttar Pradesh (46.39)
  - 2. West Bengal (22.87)
  - 3. Punbaj (5.54)
  - 4. Gujarat (5.49)
  - 5. Bihar (5.33)

- **Coconut**
- 1. Kerala (38.19) 2. Tamil Nadu (34.28)
- 3. Karnataka (10.29) 4. Andhra Pradesh (8.40)
- 5. West Bengal (2.27)

- **Fruits**

India is the largest producer of fruits in the world.

Mango: 1. Maharashtra 2. UP

Banana: 1. Maharashtra 2. Tamil Nadu

Apple: 1. J&K 2. HP

- **Milk**
- India is the largest producer in the world.

- Fish
- 1. West Bengal 2. Andhra Pradesh
- 2. Gujarat 4. Kerala
- 5. Maharashtra

- This data/ information is taken from “Directorate of Economics & Statistics”, Ministry of Agriculture, Government of India.